



Pharm*Achieve*

HYPERTENSION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ABPM Ambulatory Blood Pressure Monitoring

ACEi Angiotensin-Converting Enzyme Inhibitor

AOBP Automated Office Blood Pressure

ARB Angiotensin Receptor Blocker

ARNI Angiotensin Receptor/Neprilysin Inhibitor

BB Beta Blocker

BP Blood Pressure

CBT Cognitive Behavioral Therapy

CKD Chronic Kidney Disease

CVD Cardiovascular Disease

DBP Diastolic Blood Pressure

DHP-CCB Dihydropyridine Calcium Channel Blocker

eGFR Estimated Glomerular Filtration Rate

HBPM Home Blood Pressure Monitoring

HTN Hypertension

LA-CCB Long-Acting Calcium Channel Blocker

LVH Left Ventricular Hypertrophy

Non-DHP-CCB Non-Dihydropyridine Calcium Channel Blocker

NSAID Non-steroidal Anti-Inflammatory Drug

OBPM Office Blood Pressure Monitoring

PO By mouth

SBP Systolic Blood Pressure

SPC Single Pill Combination

THZ Thiazide

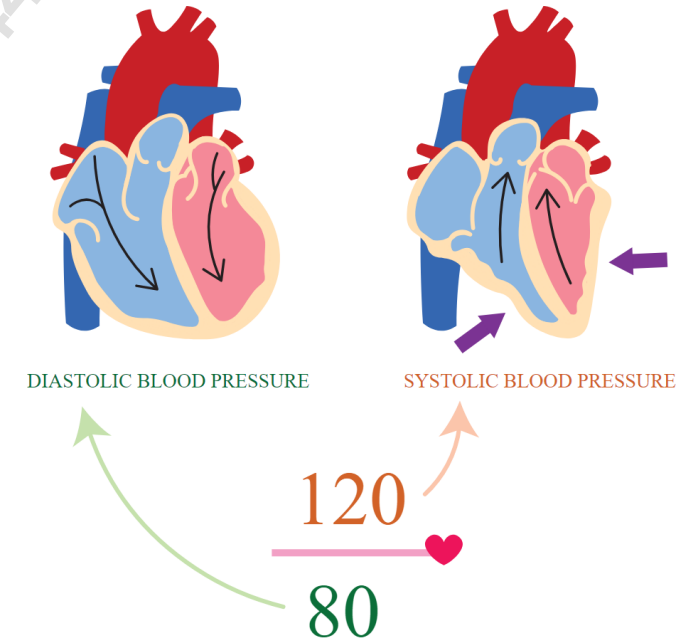
TIA Transient Ischemic Attack

TOD Target Organ Damage

INTRODUCTION

Blood pressure
pressure exerted by
blood on arterial wall

- **Diastolic BP:**
heart muscles **relax**,
↓BP as blood fills the heart
- **Systolic BP:**
heart muscles **contract**,
↑ BP as blood is pumped out of the heart



Hypertension
persistently elevated
systolic and/or
diastolic pressure

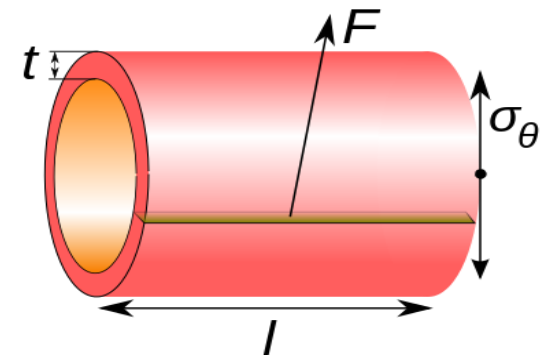
- **Hypertension:** $\geq 130/80$ mmHg
- **Isolated systolic HTN:** elevated SBP, normal DBP
- **White coat HTN:** occurs only in clinical settings (e.g. in the doctor's office)
- **Masked HTN:** normal BP in clinical settings, but elevated in non-clinical settings
- **Hypertensive emergency:** significant elevation in BP+ organ damage
- **Hypertensive urgency:** significant elevation in BP but no organ damage

PATHOPHYSIOLOGY

$$\text{BP} = \text{Cardiac Output (CO)} \times \text{Systemic Vascular Resistance (SVR)}$$

- **Cardiac output (CO):** volume of blood ejected from ventricles/minute
 - $\text{CO} = \text{HR (heart rate)} \times \text{SV (stroke volume)}$
 - **Stroke volume (SV):** volume of blood ejected during systole
 - Depends on:
 - Preload: stretching of cardiac myocytes prior to systole (determined by end-diastolic pressure)
 - Afterload: pressure that must be overcome for ejection of blood
- **Systemic Vascular Resistance (SVR):** the resistance to blood flow offered by all the systemic vasculature

Decreasing the arterial space will increase resistance with which the blood must fight against



PATHOPHYSIOLOGY

Essential hypertension believed to be caused by dysfunction in any of the following regulatory mechanisms:

Neuronal regulation

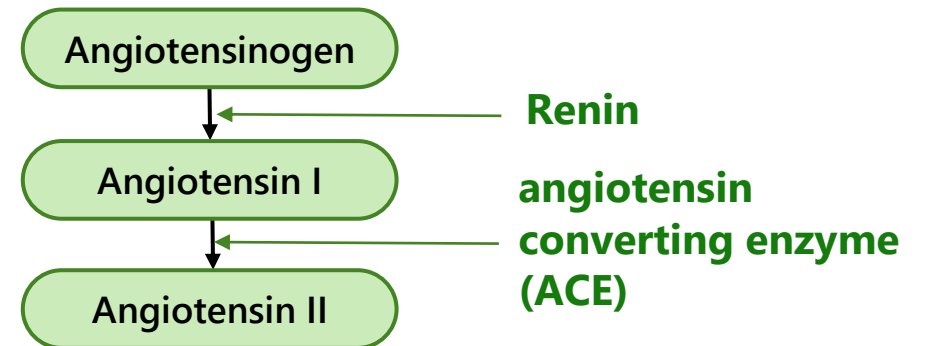
1. α_1 stimulation in periphery $\rightarrow$ vasoconstriction
2. β_1 stimulation $\rightarrow$ $\uparrow$ HR and contractility
3. Baroreceptors sense BP changes $\rightarrow$ dysfunction can result in HTN

Renin-angiotensin-aldosterone system (RAAS) activation

Kidney releases renin in response to hypotension, sympathetic stimulation, and $\downarrow$ blood flow to kidneys:

Angiotensin II:

- $\uparrow$ cardiac contractility and CO
- $\uparrow$ aldosterone synthesis $\rightarrow$ $\uparrow$ Na/H₂O
- Vasoconstriction $\rightarrow$ $\uparrow$ SVR

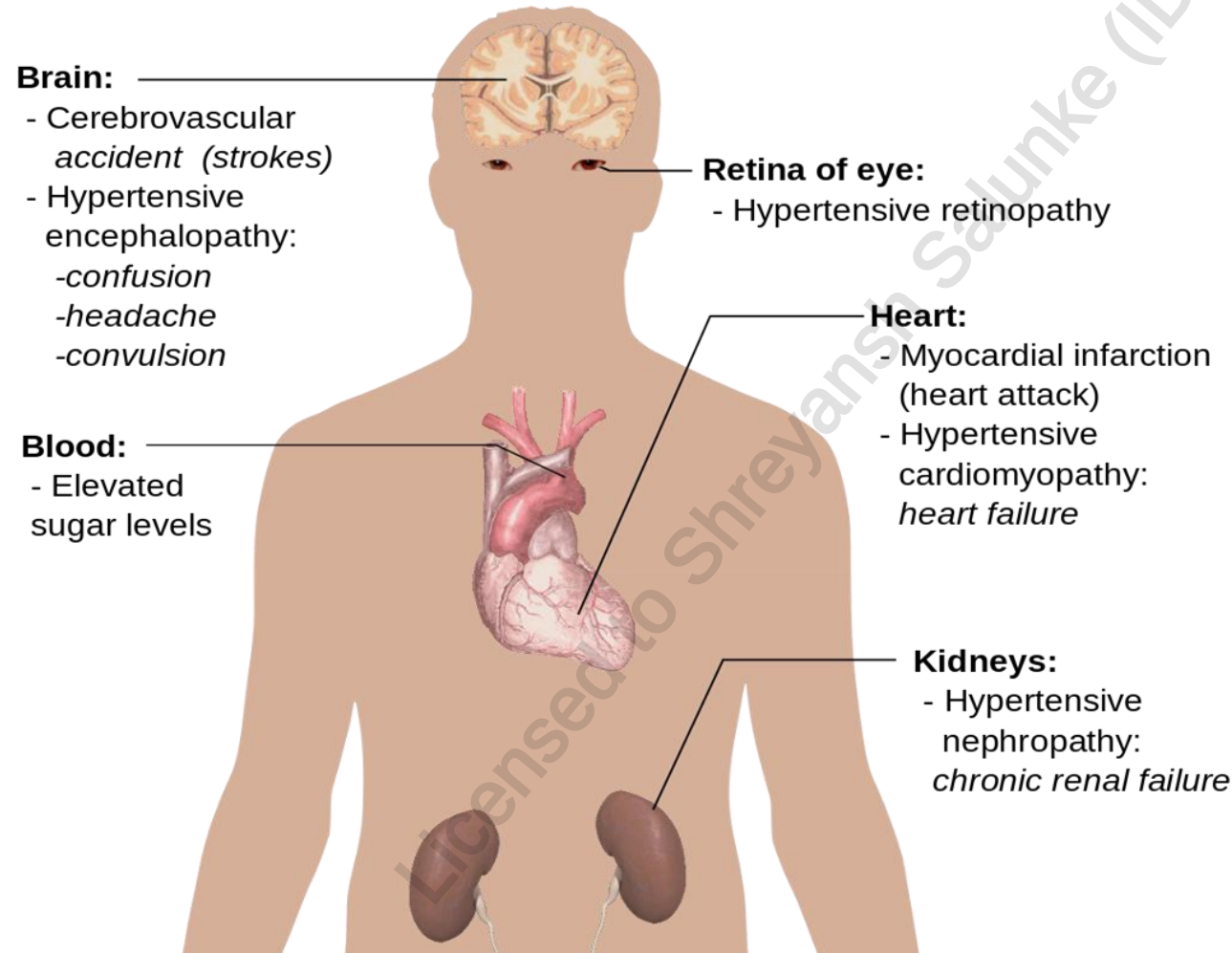


Other systems

1. Natriuretic peptide: may $\uparrow$ intracellular Na⁺ and vascular tone
2. Nitric oxide: a potent direct vasodilator
3. Electrolytes: Na⁺ = fluid retention

CLINICAL PRESENTATION

MAIN COMPLICATIONS OF PERSISTENT HIGH BLOOD PRESSURE



**HTN is usually
asymptomatic!**

CARDIOVASCULAR RISK FACTORS

NON-MODIFIABLE

- Age → ≥ 55 years
- Gender → males > females
- Race
- Family history of premature CVD (age < 55 male and < 65 female)

MODIFIABLE

- Smoking/tobacco use
- Excessive alcohol use
- Poor diet
- High sodium intake
- Sedentary lifestyle
- Mental health (i.e. depression)
- Stress
- Poor diabetes control
- Chronic pain
- Non-adherence
- Obesity
- Sleep apnea

DRUG-RELATED AND MEDICAL CAUSES

Drug-Related Causes

- Antidepressants: MAOIs, SNRIs (rare, dose dependent)
- Appetite suppressants: ephedra, caffeine
- Calcineurin inhibitors: cyclosporine, tacrolimus
- Corticosteroids (e.g. prednisone) /NSAIDs/anabolic steroids
- Depressant/nicotine withdrawal
- Erythropoietin and analogues
- Excessive alcohol intake
- Excess thyroid hormone: exogenous or endogenous
- Licorice (black)
- Midodrine
- Oral contraceptives, sex hormones (estrogen)
- Stimulants: cocaine, nicotine, amphetamines, sympathomimetic agents
- Atypical antipsychotics
- Decongestants
- Tyrosine Kinase inhibitors
- St. John's wort

Medical Causes

- Kidney disease
- Thyroid disease
- Pheochromocytoma
- Hyperaldosteronism
- Sleep apnea
- Obesity
- Cushing's Syndrome
- Parathyroid Disease

HYPERTENSION IS A RISK FACTOR FOR MANY CONDITIONS

- Transient ischemic attack (TIA)/ Stroke
- Myocardial infarction/Angina
- Congestive heart failure
- Chronic kidney disease
- Peripheral vascular disease (i.e. intermittent claudication)
- Dementia
- Atrial fibrillation
- Erectile dysfunction
- Retinopathy

BP MEASURING

BP should be taken using a validated automated device using a standardized protocol

Proper BP measurement tips:

- Rest in a **quiet** place for **1-5 minutes** before measurements
- **Sit** in a chair with **back supported**
- **Feet flat on the ground**, not crossed
- **No caffeine, tobacco or exercise** in the past 30 minutes (e.g. running to the doctor's office)
- **Ensure cuff size fits the arm** (length of cuff should cover 80-100% of the arm circumference)
- Arm should be bare, **supported and kept at heart level**
- Take **3 BP readings** at 1-minute intervals and record and average all values



***DO NOT** measure if:

- Stressed, cold, in pain
- Uncomfortable
bowel/bladder

GOALS OF THERAPY

- ✓ Control/eliminate modifiable risk factors
- ✓ Prevent complications and morbidity associated with HTN
- ✓ Provide convenient and cost-effective therapy to improve adherence
- ✓ Maximize patient adherence to reach target BP in an adequate time frame
- ✓ Treat to target BP

HYPERTENSION CANADA

PATIENT STRATIFICATION

Hypertension Canada stratifies patients by cardiovascular risk, and based on that risk, there are different thresholds for initiating treatment

High cardiovascular disease risk criteria:

- ☐ Established cardiovascular disease (coronary artery disease, heart failure, cerebrovascular disease, and peripheral artery disease)
- ☐ Diabetes mellitus (type 1 or type 2)
- ☐ Chronic kidney disease (eGFR < 60 mL/min/1.73 m² or albuminuria ≥ 3 mg/mmol)
- ☐ 10-year Framingham Risk Score ≥ 20%
- ☐ Age ≥ 75 years

SPRINT TRIAL TAKEAWAY

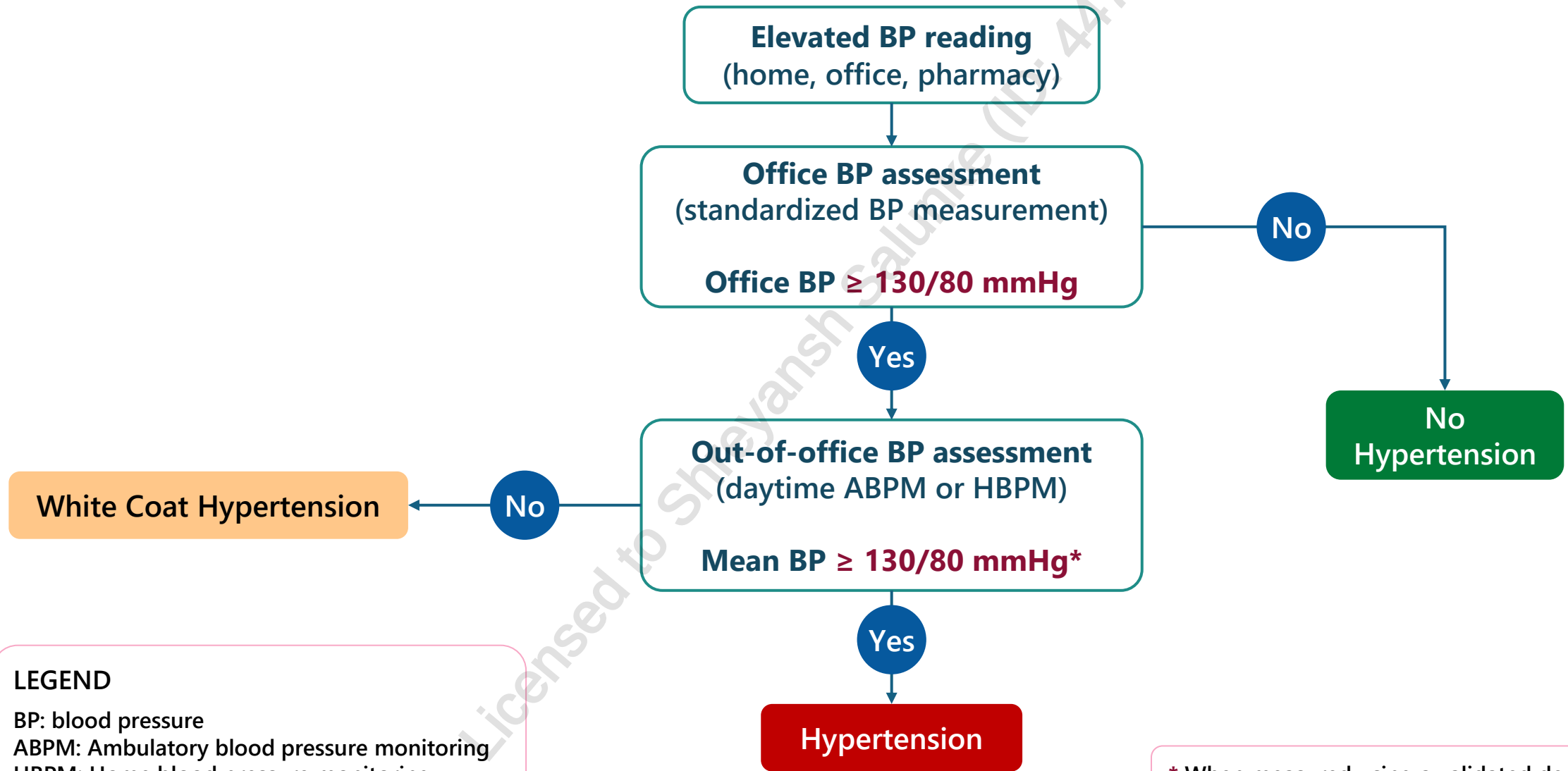
- **Objective:** To determine whether intensive systolic BP control (<120 mmHg) reduces cardiovascular (CV) events compared to standard control (<140 mmHg).
- **Participants:** 9,361 adults (≥ 50 years) with systolic BP ≥ 130 mmHg and increased CV risk (without diabetes or history of stroke).
- **Groups:**
 - Intensive Treatment: Target SBP <120mmHg
 - Standard Treatment: Target SBP <140mmHg
- **Results/Conclusion:**
 - Intensive treatment reduced rates of CV events (heart attack, stroke, HF) and all-cause mortality.
 - 25% lower risk of CV events
 - 27% lower risk of death
 - Increased rates of hypotension, syncope, electrolyte abnormalities, and AKI.

TARGET ORGAN DAMAGE

TOD should be assessed in any person with hypertension

1. Cardiovascular disease (CAD, angina, HF, LVH)
2. Cerebrovascular disease (ischemic stroke or TIA, dementia, intracerebral hemorrhage)
3. Peripheral arterial disease
4. Hypertensive retinopathy
5. Renal disease

DIAGNOSIS



LEGEND

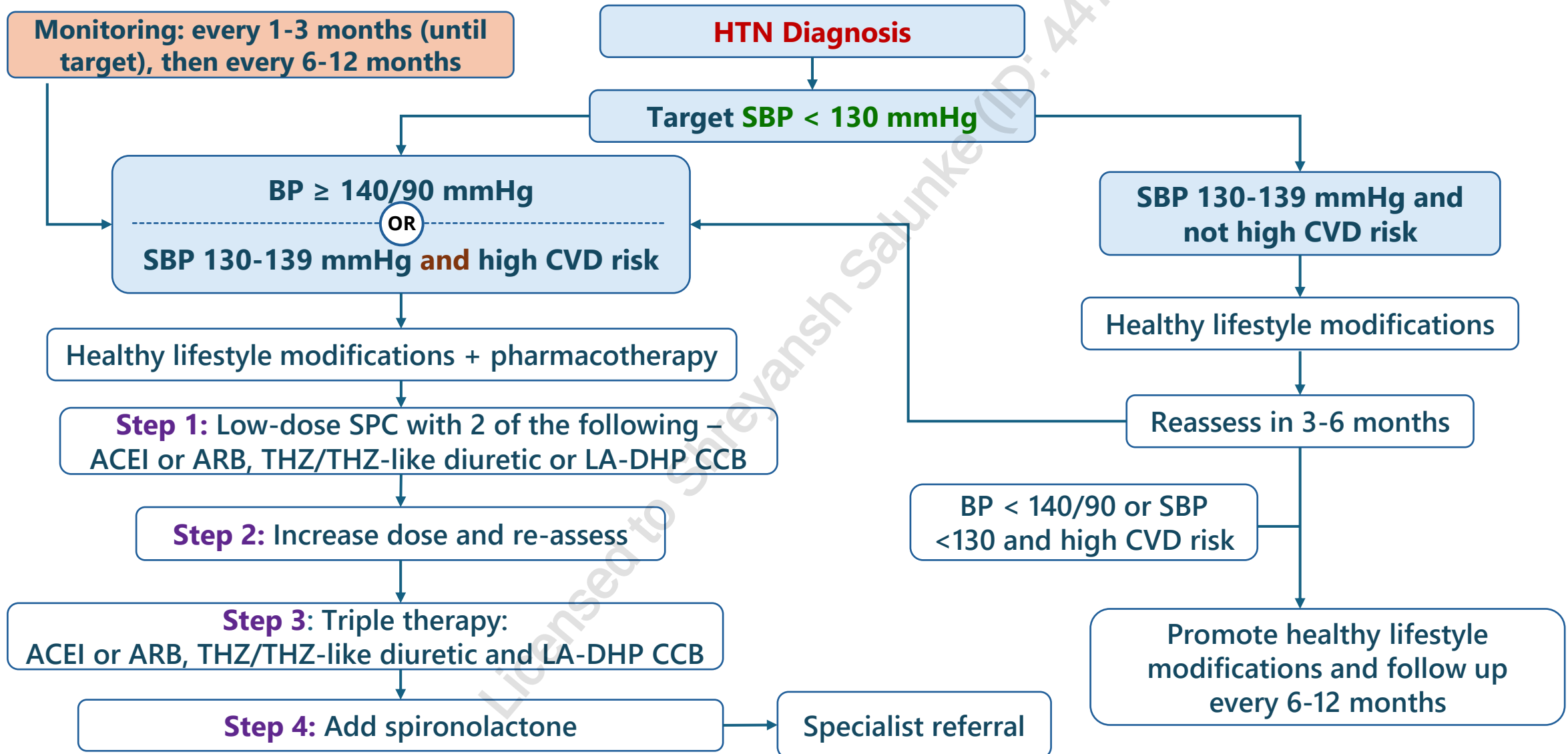
BP: blood pressure

ABPM: Ambulatory blood pressure monitoring

HBPM: Home blood pressure monitoring

* When measured using a validated device under optimal conditions.

TREATMENT ALGORITHM



NON-PHARMACOLOGICAL MEASURES

- **Weight control** (BMI 18.5-24.9 and waist circumference <102 cm for men <88 cm for women) to prevent HTN and to reduce BP in hypertensive patients
- **Smoking cessation**
- DASH-like diet (fruits, vegetables, low-fat dairy, dietary fibre, non-animal protein, low in saturated fat)
- Reduction in alcohol consumption – lowers BP in a dose-dependent fashion
- **Physical activity** (at least 150-300 minutes of moderate aerobic activity weekly)
- **Reduction in sodium intake** (< 2g/ day)
- Increased dietary potassium if not at risk for hyperkalemia (i.e. fresh fruit, vegetables, legumes)
- Relaxation therapies: CBT and relaxation techniques for patients where stress plays a role in elevating BP

PHARMACOLOGICAL MEASURES

SCENARIO

Hypertension without other compelling indications

Health Behaviour Management, and....

If pharmacotherapy is indicated -

Any 2 of the following in a SPC, as initial therapy:

- **Thiazide/thiazide-like diuretics:** long-acting, such as indapamide and chlorthalidone, are preferred over shorter-acting like hydrochlorothiazide
- **LA-DHP CCBs:** consider risk of peripheral edema
- **ACEIs:** caution in females of child-bearing potential (teratogen) and patients with hyperkalemia; not recommended as monotherapy in persons of African descent unless concomitant diabetes
- **ARBs:** caution in females of child-bearing potential (teratogen) and patients with hyperkalemia
→ Do NOT combine ACEi and ARBs!

Note: **BBs** have a similar degree of BP reduction, but the magnitude of reduction of CVD risk is less. Also, discontinuation rates are higher due to adverse reactions. NOT recommended as 1st line for hypertension except in the presence of compelling indications, e.g. HF, angina, post-MI, heart rate or rhythm control.

SPECIAL POPULATIONS AND CONSIDERATIONS

RECOMMENDATIONS ARE BASED ON
CURRENT 2020 GUIDELINES, PENDING
THE RELEASE OF THE 2025 UPDATES

SCENARIO

Adults with diabetes mellitus **PLUS** one of:
microalbuminuria, renal disease, CVD, or other CV risk factors

- *1st line therapy:* **ACEi or ARB:** caution in females of child-bearing potential (**teratogen**), and patients with risk factors for hyperkalemia
- *Second-line therapy:* **Addition of DHP-CCB** preferred over thiazide/thiazide-like diuretic

SCENARIO

Adults with diabetes mellitus **WITHOUT** any of the risk
factors in the scenario above

- *1st line therapy:* **ACEi or ARB or DHP-CCB or thiazide/thiazide-like diuretics**
- *Second-line therapy:* Combination of 1st line → if combination with ACEi is required, addition of **DHP-CCB is preferred**

SPECIAL POPULATIONS AND CONSIDERATIONS

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THE RELEASE OF THE 2025 UPDATE

Persons of African Descent

- NOT recommended: ACEi/ARB unless other compelling indication, e.g. kidney protection, heart failure or diabetes

Pregnancy

- *1st line:* Methyldopa, labetalol, long-acting nifedipine, other BB (acebutolol, metoprolol, pindolol, propranolol) may also be considered
- *2nd line:* hydralazine, clonidine, thiazide diuretics
- Pre-eclampsia and eclampsia: IV magnesium, IV labetalol, IV hydralazine
- NOT recommended: ACEi or ARBs
- Severe HTN (BP $\geq$ 160/110 mmHg) is considered an obstetrical emergency
- Target BP: DBP

SPECIAL POPULATIONS AND CONSIDERATIONS

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Coronary Artery Disease (angina, MI)

- 1st line: BBs and ACEi (or ARBs if ACEi not tolerated)
- *2nd line:* Add LA-DHP CCB

Heart Failure

- *1st line:* BBs and ACEi (or ARBs if ACEi intolerant)
 - Aldosterone antagonists may be added for non-HTN reasons
 - In HFrEF, ARNI is preferred over an ACEi or ARB
- *2nd line:* hydralazine/ISDN combo if ACEi is contraindicated or not tolerated; thiazide or loop diuretic as additive therapy; DHP-CCB; combo ARB/neprilysin-inhibitor

SPECIAL POPULATIONS AND CONSIDERATIONS

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THE RELEASE OF THE 2025 UPDATE

Past Stroke or TIA

- *1st line:* ACEi + THZ diuretic

Non-diabetic CKD with Proteinuria

- *1st line:* ACEi (or ARBs if ACEi intolerant) if proteinuria is present. Diuretics as additive therapy

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
BP	SBP < 130 mmHg (generally)	Every 1-3 months until at target, then every 6-12 months

Safety Endpoints		
Parameter	Degree of Change	Timeframe
ADR from medication therapy e.g. dizziness, dry cough, peripheral edema...etc.	Minimal to none	Duration of therapy
K+ levels	Within target	2-4 weeks after initiation and with any dose adjustment (e.g. with spironolactone)
SCr levels	Within target	Depending on the clinical situation

POSSIBLE REASONS FOR POOR RESPONSE TO THERAPY

- Inaccurate BP measurement
- Suboptimal treatment regimen
- Adherence
- Associated conditions
(i.e. obesity, tobacco use, excessive alcohol use, sleep apnea, mental health)
- Drug interactions

TAKEAWAY

According to the 2025 HYPERTENSION CANADA GUIDELINES:

- Hypertension: $\geq 130/80$ mmHg
- Treatment with healthy lifestyle modifications may be enough if the SBP is 130-139 mmHg and the patient is **not** at high risk of CVD
- Pharmacological treatment is recommended in all patients with BP $\geq 140/90$ mmHg, as well as in those with SBP 130-139 mmHg who **are** at high risk of CVD
 - Pharmacotherapy should always be used in combination with lifestyle modifications
- Initial BP treatment should include 2 of the following drug classes in a SPC:
 1. ACEI or ARB
 2. THZ/THZ-like Diuretic
 3. LA-DHP CCB
- Treatment target: SBP < 130 mmHg

CASE QUESTIONS

HT is a 36-year-old female who visits your pharmacy to have her blood pressure checked. It is 202/125 mmHg, so you send her to the emergency department immediately. HT was recently laid off from her job and has been quite stressed since. HT is not currently pregnant or breastfeeding, but she and her partner are trying to conceive. HT has a medical history of hypothyroidism and migraines for which she takes levothyroxine 112 mcg PO QAM, multivitamin 1 tab PO daily, Advil® (ibuprofen) 400 mg PO PRN for headaches, and zolmitriptan 5 mg PO PRN for migraines. She smokes 5 cigarettes/day, rarely drinks alcohol, and exercises on the weekends. She is not allergic to anything.

Objective findings:

BP=202/125 mmHg, HR=92 bpm, sinus rhythm

SCr=74 µmol/L

TSH normal

Weight=85 kg, height = 5 feet, 4 inches

All other lab values and physical findings are normal

1. Which of the following is HT most likely experiencing?

- a) Masked hypertension
- b) Hypertensive urgency
- c) Hypertensive emergency
- d) Isolated systolic hypertension



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2. Which of the following is a modifiable risk factor for hypertension that HT has?

- a) Female gender
- b) Frequent use of ibuprofen
- c) Smoking status
- d) Alcohol consumption



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All other lab values and physical findings are normal

3. After HT's BP is stabilized at the hospital, which of the following medications will most likely be prescribed?

- a) Labetalol
- b) Candesartan
- c) Ramipril
- d) Nifedipine immediate release



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TSH normal

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4. HT is now pregnant and her BP is 167/110 mmHg. Which of the following is the most appropriate option for her?
- a) Add olmesartan to existing therapy
 - b) Add clonidine
 - c) Increase labetalol dose
 - d) Refer to the nearest emergency department



REFERENCES

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3. Wright JT Jr, Williamson JD, Whelton PK, et al. A Randomized Trial of Intensive versus Standard Blood-Pressure Control. *N Engl J Med*. 2015;373(22):2103-2116.
4. Goupil R, Tsuyuki RT, Santesso N, et al. Hypertension Canada guideline for the diagnosis and treatment of hypertension in adults in primary care. *CMAJ*. 2025;197(20):E549-E564. Published 2025 May 25. doi:10.1503/cmaj.241770.

CHANGE LOG

May 2025

- Slide 24: Updated information on SPC and monotherapy

July 2025

- Significant changes made according to the 2025 Hypertension Canada guidelines

Sept 2025

- Slide 14 – added stroke history as part of exclusion criteria
- All other content reviewed, no changes made

January 2026

- Slide 2: Added new NAPRA competencies
- Slide 5: Changed from 'diastole' to 'systole' in the definition of preload
- Slide 24: Reformatted monitoring parameters slide